EXCELLENCE

Class 11 Physics · Session 2026–27

Basic Maths & Calculus

Complete formula notes — Lectures 01 to 18

Full coverage of the Basic Maths block: trigonometry tables, identities, ASTC, compound angles, AP–GP, graphs, differentiation and integration. Every formula is on its own line. Symbols are explained before use.

How to use each page
1. Read “In plain English”
2. Read “What the words mean”
3. Copy each formula line separately
4. Try the green example

Full chapter map

0. Symbol dictionary
1. Trig ratios and height–distance
2. Full trig tables (0°–90° and 90°–180°)
3. Identities, range, brother / allied angles
4. ASTC, radian, arc length
5. Compound, double, half and small angles
6. Max–min of a sin + b cos (phasor)
7. AP and GP (with physics uses)
8. Coordinates, line, graph shapes, circle, ellipse
9. Differentiation — full rule set
10. Applications: motion, related rates, max–min
11. Integration — indefinite, definite, area, average
12. Revision lock lines

0. Symbol dictionary — read this first

Symbol / short name	Full meaning in school language
θ (theta)	an angle (degree or radian — check the question)
P, B, H	opposite side, adjacent side, hypotenuse
sin, cos, tan	sine, cosine, tangent
cosec, sec, cot	1/sin, 1/cos, 1/tan
ND	not defined (division by zero)
rad	radian — angle unit used in calculus and arc = Rθ
π (pi)	≈ 3.14 ; also π radian = 180°
s, R	arc length, radius
a, d, n (AP)	first term, common difference, number of terms
a, r, n (GP)	first term, common ratio, number of terms
m, c	slope of a line, y-intercept
dy/dx , y' , f'(x)	derivative = rate of change / tangent slope
d^2y/dx^2	second derivative (used for max / min test)
∫ f(x) dx	indefinite integral (anti-derivative) + C
$\int_a^b f(x)\,dx$	definite integral = F(b) − F(a) (no +C)
ω, φ	angular frequency, phase angle

Remember: If a formula uses a letter you forgot, open this page first.

1. Trig ratios and height–distance

In plain English: pick an acute angle θ in a right triangle. Name opposite (P), adjacent (B) and hypotenuse (H). Six ratios are fractions from these.

What the words mean
- Opposite (P) = side facing θ
- Adjacent (B) = side next to θ (not hypotenuse)
- Hypotenuse (H) = longest side
- Angle of elevation = looking up from horizontal
- Angle of depression = looking down from horizontal

Name	Write it as	Also equal to
sine (sin θ)	P / H	1 / cosec θ
cosine (cos θ)	B / H	1 / sec θ
tangent (tan θ)	P / B	sin θ / cos θ
cosecant (cosec θ)	H / P	1 / sin θ
secant (sec θ)	H / B	1 / cos θ
cotangent (cot θ)	B / P	1 / tan θ

sin θ × cosec θ = 1
cos θ × sec θ = 1
tan θ × cot θ = 1

Height and distance (right triangle)
height = (distance along ground) × tan(elevation)
If elevation from A is α and from B is β (same tower),
use two tan equations and solve for height

Remember: Tan is never P/H. Tan is always P/B. Elevation and depression are measured from the horizontal.

Worked example — find other ratios
• cos θ = 8/17 (acute). Adjacent = 8, hypotenuse = 17.
• Opposite = $\sqrt{(17^2 - 8^2)} = 15$.
• sin θ = 15/17 , tan θ = 15/8.
Final answer: sin θ = 15/17 , tan θ = 15/8

2. Full trig tables — do not skip this page

In plain English: learn the pattern. ND means not defined. The table is split into two parts so every cell prints clearly on the page.

Table A — 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	ND
cosec	ND	2	√2	2/√3	1
sec	1	2/√3	√2	2	ND
cot	ND	√3	1	1/√3	0

Table B — 90° to 180°

θ	90°	120°	135°	150°	180°
sin	1	√3/2	1/√2	1/2	0
cos	0	−1/2	−1/√2	−√3/2	−1
tan	ND	−√3	−1	−1/√3	0
cosec	1	2/√3	√2	2	ND
sec	ND	−2	−√2	−2/√3	−1
cot	0	−1/√3	−1	−√3	ND

Brother angles: 120° ↔ 60°, 135° ↔ 45°, 150° ↔ 30° (from 180°). Physics favourites: sin 37° ≈ 3/5, cos 37° ≈ 4/5; sin 53° ≈ 4/5, cos 53° ≈ 3/5.

Beyond 180°: use ASTC for the sign, then reduce to an acute angle
Example idea: sin(210°) = sin(180°+30°) = −sin 30° = −1/2

Worked example — read the table
• sin 150° = 1/2 , sec 120° = −2.
Final answer: sin 150° = 1/2 , sec 120° = −2

3. Identities, range, brother / allied angles

In plain English: identities are always true (where defined). Start from $\sin^2\theta + \cos^2\theta = 1$. The other two are the same line divided by $\cos^2\theta$ or $\sin^2\theta$.

What the words mean
• $\sin^2\theta$ means $(\sin\theta)\times(\sin\theta)$, not $\sin(\sin\theta)$
• Brother / allied angle = rewrite using 90°±θ, 180°±θ, 270°±θ, 360°±θ
• At 90° and 270°, sin ↔ cos (function may change). At 180° and 360°, function stays, sign may change

$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Range: $-1 \le \sin\theta \le 1$
Range: $-1 \le \cos\theta \le 1$
$\tan\theta$ can be any real number (where defined)

Useful brother lines

$\sin(90° - \theta) = \cos\theta$

$\cos(90° - \theta) = \sin\theta$

$\sin(180° - \theta) = \sin\theta$

$\cos(180° - \theta) = -\cos\theta$

$\sin(180° + \theta) = -\sin\theta$

$\cos(180° + \theta) = -\cos\theta$

More brother lines

$\sin(360° - \theta) = -\sin\theta$

$\cos(360° - \theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$ (odd)

$\cos(-\theta) = \cos\theta$ (even)

$\tan(-\theta) = -\tan\theta$ (odd)

$\tan(180° + \theta) = \tan\theta$

4. ASTC, radian and arc length

In plain English: ASTC tells which ratios stay positive in each quadrant (anti-clockwise from the first). Radian is another angle unit; physics needs it for arc and calculus.

What the words mean
- 1st quadrant (0°–90°): All positive (A)
- 2nd (90°–180°): only Sin / cosec positive (S)
- 3rd (180°–270°): only Tan / cot positive (T)
- 4th (270°–360°): only Cos / sec positive (C)
- s = arc length, R = radius, θ in radian for $s = R\theta$

ASTC (anti-clockwise from 1st quadrant)
A = All positive
S = only Sin positive
T = only Tan positive
C = only Cos positive

π radian = 180°
degree → radian: multiply by $\pi/180$
radian → degree: multiply by $180/\pi$
Arc length: $s = R \times \theta$ (θ in radian)

Remember: For differentiation / integration of sin, cos, tan — angle must be in radian. Example: π/3 rad = 60°.

Worked example — radian to degree
- $(5\pi/6) \times (180°/\pi) = 150°$.

Final answer: 150°

5. Compound, double, half and small angles

In plain English: expand sin/cos/tan of a sum or difference. Double angle is the same with both angles equal. Half angle and small-angle shortcuts are below.

Sine and cosine of A ± B

sin(A+B) = sin A cos B + cos A sin B

sin(A−B) = sin A cos B − cos A sin B

cos(A+B) = cos A cos B − sin A sin B

cos(A−B) = cos A cos B + sin A sin B

Tangent of A ± B

tan(A+B) = (tan A + tan B)/(1 − tan A tan B)

tan(A−B) = (tan A − tan B)/(1 + tan A tan B)

If tan A × tan B = 1, tan(A+B) is ND

$\sin 2\theta = 2 \sin\theta \cos\theta$
$\cos 2\theta = \cos^2\theta - \sin^2\theta$
$\cos 2\theta = 2\cos^2\theta - 1$
$\cos 2\theta = 1 - 2\sin^2\theta$
$\tan 2\theta = 2\tan\theta / (1 - \tan^2\theta)$

$\sin^2\theta = (1 - \cos 2\theta)/2$
$\cos^2\theta = (1 + \cos 2\theta)/2$
sin(θ/2) and cos(θ/2) use half-angle forms from these

Small angle (θ in radian, about 5° or less)
$\sin\theta \approx \theta$
$\tan\theta \approx \theta$
$\cos\theta \approx 1$
Never put degrees inside this shortcut

Worked example — compound angle
• sin 75° = sin(45°+30°) = sin45 cos30 + cos45 sin30
• = (√6 + √2)/4.
Final answer: (√6 + √2)/4

6. Max–min of a sin θ + b cos θ (phasor)

In plain English: pack a sine and a cosine into one sine with amplitude R. Used in SHM, waves and AC.

What the words mean
• a, b = fixed numbers
• R = √(a² + b²) = biggest size of the sum
• φ = phase; tan φ = b/a (choose correct quadrant)

$y = a\sin\theta + b\cos\theta$
Maximum of $y = \sqrt{(a^2 + b^2)}$
Minimum of $y = -\sqrt{(a^2 + b^2)}$

$a\sin\omega t + b\cos\omega t = R\sin(\omega t + \varphi)$
$R = \sqrt{(a^2 + b^2)}$
$\tan\varphi = b / a$

7. AP and GP

In plain English: AP = equal adding. GP = equal multiplying.

What the words mean
- AP: a = first term, d = common difference, n = number of terms
- GP: a = first term, r = common ratio
- T_n = n-th term, S_n = sum of n terms, S_∞ = endless sum ($|r|<1$ only)

Arithmetic progression (AP)

$T_n = a + (n-1)d$

$S_n = n/2\ [2a + (n-1)d]$

$S_n = n/2$ (first + last)

$1+2+\ldots+n = n(n+1)/2$

Geometric progression (GP)

$T_n = a\ r^{n-1}$

$S_n = a(r^n - 1)/(r - 1)$ $(r \neq 1)$

$S_\infty = a/(1-r)$ only if $|r|<1$

Physics: repeating resistors, decaying series

Remember: Do not use S_∞ when $|r| \geq 1$. The endless sum does not settle.

8. Coordinates, line, graph shapes, circle, ellipse

In plain English: graphs are stories on paper. Always label quantity and unit on each axis.

Distance between (x_1,y_1) and (x_2,y_2)
$$\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Slope $m = (y_2 - y_1)/(x_2 - x_1) = \Delta y/\Delta x = \tan\theta$
Straight line: $y = mx + c$
Parallel lines: same m
Perpendicular lines: $m_1 m_2 = -1$

Shapes to recognise

Modulus: $y = |x|$ (V-shape)

Hyperbola: $y = k/x$ or $xy = k$

Parabola: $y = ax^2 + bx + c$

Square root: $y = \sqrt{x}$

Exponential: $y = e^x$ or e^{-x}

Physics graph links

$F = -kx \rightarrow$ straight line through origin (3rd/1st)

$K \propto v^2 \rightarrow$ parabola in v

$T \propto \sqrt{L} \rightarrow$ square-root shape

Ideal gas: pV = constant → hyperbola

v–t area = displacement

Circle

$x^2 + y^2 = R^2$ (centre origin)

$(x-h)^2 + (y-k)^2 = R^2$

Ellipse / SHM

$x^2/a^2 + y^2/b^2 = 1$

SHM: $|v| = \omega\sqrt{(A^2 - x^2)}$

Quadratic: $a\,x^2 + b\,x + c = 0$
$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$
Discriminant $D = b^2 - 4ac$
$D>0$ two real roots; $D=0$ one; $D<0$ no real root

9. Differentiation — full rule set

In plain English: differentiation finds how fast y changes when x changes a tiny bit. On a graph it is the tangent slope.

What the words mean
• dy/dx = derivative of y with respect to x
• Also written y' or f'(x)
• Trig rules need the angle in radian

dy/dx = limit of (Δy/Δx) as Δx → 0

If y is…	Then dy/dx is…	In plain words
constant C	0	Flat — no change
x^n	$n\,x^{n-1}$	Power rule
k × f(x)	k × f'(x)	Constant walks out
A + B	A' + B'	Term by term
A − B	A' − B'	Term by term
e^x	e^x	Stays the same
ln|x|	1/x	Natural log
sin x	cos x	x in radian
cos x	−sin x	x in radian
tan x	$\sec^2 x$	Where defined
sec x	sec x tan x	
cosec x	−cosec x cot x	
cot x	$-\mathrm{cosec}^2 x$	

Product rule: (AB)' = A B' + B A'
Quotient rule: (A/B)' = (B A' − A B') / B^2
Chain rule: dy/dx = (dy/du)×(du/dx)
If $y = e^u$ then $dy/dx = e^u \times (du/dx)$
If y = sin u then dy/dx = cos u × (du/dx)

Parametric: if x=x(t), y=y(t)
dy/dx = (dy/dt) / (dx/dt)
Partial derivative: vary one variable, hold others fixed
Example: ∂V/∂x means differentiate V w.r.t. x only

10. Applications — motion, related rates, max–min

In plain English: the same derivative idea appears in motion and in “how fast is this changing”.

v = dx/dt
a = dv/dt = d^2x/dt^2
Related rates: differentiate both sides with respect to time

Stationary point: dy/dx = 0
If $d^2y/dx^2 > 0$ → local minimum (valley)
If $d^2y/dx^2 < 0$ → local maximum (hill)
If $d^2y/dx^2 = 0$ → test further (may be inflection)

Worked example — related rates idea
• Circle area $A = \pi r^2$. Differentiate w.r.t. time t:
• $dA/dt = 2\pi r\,(dr/dt)$.
Final answer: $dA/dt = 2\pi r\,(dr/dt)$

11. Integration — indefinite, definite, area, average

In plain English: integration undoes differentiation. Indefinite integral has +C. Definite integral has limits and gives a number (no +C).

What the words mean
- F(x) = anti-derivative of f (so F' = f)
- C = constant of integration
- a = lower limit, b = upper limit

$\int f(x)\,dx = F(x) + C$
$\int x^n\,dx = x^{n+1}/(n+1) + C \ (n \neq -1)$
$\int_a^b f(x)\,dx = F(b) - F(a)$
(no +C in a definite integral)

Function f(x)	∫ f(x) dx
k (constant)	kx + C
x^n ($n \neq -1$)	$x^{n+1}/(n+1) + C$
1/x	ln|x| + C
e^x	$e^x + C$
sin x	−cos x + C
cos x	sin x + C
$\sec^2 x$	tan x + C
sec x tan x	sec x + C
$\text{cosec}^2 x$	−cot x + C

Area under curve (above axis) = $\int_a^b f(x)\,dx$
Area between curves = ∫ (upper − lower) dx
Average value = $(1/(b-a)) \times \int_a^b f(x)\,dx$
Displacement = area under v–t graph
$v = \int a\,dt + C_1$, $x = \int v\,dt + C_2$

Substitution (chain rule backwards)
1. Choose u = inside piece
2. Write du = (…) dx
3. Rewrite the whole integral in u
4. For definite integrals, change both limits to u values

Worked example — power rule
- $\int 5x^2\,dx = 5 \times (x^3/3) + C = (5/3)x^3 + C$.
- Check: differentiate to get $5x^2$ back.

Final answer: $(5/3)\,x^3 + C$

12. Revision lock lines — say these aloud

Topic	Say this in one breath
Ratios	sin=P/H, cos=B/H, tan=P/B. Tan is never P/H.
Table	Know 0°–180° for all six ratios. ND = not defined.
Identity	$\sin^2\theta + \cos^2\theta = 1$; then $1+\tan^2\theta=\sec^2\theta$.
ASTC	All, Sin, Tan, Cos — positive in that order around the circle.

Topic	Say this in one breath
Radian	$s = R\theta$ with θ in radian; π rad = 180°.
Double	$\sin 2\theta = 2\sin\theta\cos\theta$; store three forms of $\cos 2\theta$.
Small angle	$\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1$ (θ in radian).
a sin+b cos	Biggest value $\sqrt{(a^2+b^2)}$.
AP / GP	Equal add vs equal multiply; S_∞ needs $\lvert r\rvert<1$.
Line	$y=mx+c$; perpendicular slopes multiply to -1.
Differentiate	Power, product, quotient, chain.
Max / min	$y'=0$; $y''>0$ valley; $y''<0$ hill.
Integrate	Undo derivative; definite = $F(b)-F(a)$.
Area / average	Area = definite integral; average = that ÷ $(b-a)$.

Last tip

Keep the printed trig tables on your desk while you revise.
Use radian for calculus trig. Label graph axes with units.
If a formula is not on this sheet but was in class, add it by hand on the blank margin.

Source: Excellence Basic Maths & Calculus lectures 01–18. For revision — not a question paper.